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# Catch Your Breath: Adaptive Computation for Self-Paced Sequence Production

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## Abstract

We explore a class of supervised training objectives that allow a language model to dynamically and autonomously scale the number of compute steps used for each input token. For any token, the model can request additional compute steps by emitting a <DON'T KNOW> output. If the model is granted a delay, a specialized <PAUSE> token is inserted at the next input step, providing the model with additional compute resources to generate an output. The model can request multiple pauses. To train the model to use <DON'T KNOW> outputs judiciously and to calibrate its uncertainty, we frame the selection of each output token as a sequential-decision problem with a time cost. We refer to the class of methods as *Catch Your Breath* losses and we study three methods in this class: CYB-AP frames the model's task as anytime prediction, where an output may be required at any step and accuracy is discounted over time; CYB-VA is a variational approach that aims to maximize prediction accuracy subject to a specified distribution over stopping times; and CYB-DP imposes a penalty based on a computational budget. Through fine-tuning experiments, we determine a specific form of the loss that performs best. To cast the performance improvement in intuitive terms, a baseline (no pause) model needs $3\times$ as much training data to match the CYB loss, and a model with pauses and a cross-entropy loss requires $2\times$ as much data. We find that the CYB model requests additional steps when doing so improves accuracy, and the model adapts its processing time to token-level complexity and context. For example, it often pauses after plural nouns like *patients* and *challenges* but never pauses after the first token of contracted words like *wasn* and *didn*, and it shows high variability for ambiguous tokens like *won*, which could function as either a verb or part of a contraction.

## 1 Introduction

The transformer architecture (Vaswani et al., 2018) can be scaled along three distinct dimensions: dataset size, model size, and amount of computation. Rapid progress in foundational models was initially achieved by focusing on increasing the size of data sets and models. Recently, the field's focus has shifted to computational scaling—varying the amount of computation time to produce a response. Most of this effort involves collecting multiple samples from a model or extending reasoning chains via prompting (e.g., Zelikman et al., 2022; Wei et al., 2023; Yao et al., 2023) or enabling latent thought (e.g., Phan et al., 2023; Pfau et al., 2024; Hao et al., 2024; Ruan et al., 2025). However, benefits of computational scaling can also be attained by increasing the resources devoted

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to individual tokens, such as dynamically varying the model depth (Raposo et al., 2024) or the number of layer iterations at inference time (Chen et al., 2025; Geiping et al., 2025).

In this article, we explore an approach to dynamically scaling the number of compute steps for individual tokens by training the model to selectively request additional compute. As the input is processed step by step, the model is given the option of emitting a special <DON'T KNOW> token which requests a delay. If the delay is granted by the data loader, a <PAUSE> token is inserted in the input stream (Goyal et al., 2024), holding off on the arrival of further input and allowing the model an additional forward pass to process the sequence to that point (Figure 1). The model can produce the <DON'T KNOW> response multiple times in a row, though we design training objectives to ensure that the model will not abuse this privilege.

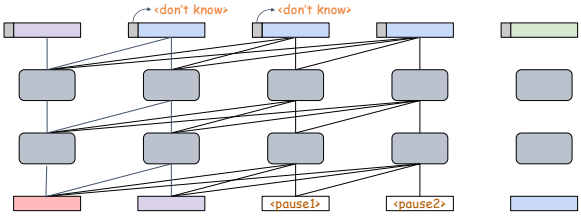


Figure 1: Sequence processing in a transformer with <DON'T KNOW> outputs and <PAUSE> inputs.

### 1.1 How People Read

The inspiration for this research comes from studies of human reading. When gaze is tracked as an individual reads, inter-saccade durations are highly nonuniform. Readers fixate longer when processing demands rise, e.g., at points where information can be integrated across phrases, where inferences can be made at the end of sentences, and where low-frequency terms occur (Just and Carpenter, 1980). Figure 2 shows fixation order within a sentence for a human reader as well as the fixation duration. Note that short function words are skipped or quickly glanced over. When an uncommon interpretation of a word or sentence is required, readers pause. For instance, when idioms that typically have a figurative meaning (e.g., 'break the ice') are used in a literal way, processing time slows at the point where the disambiguating information is provided (Arnon and Lavidor, 2023). Similarly, readers pause when anomalous or unexpected information is provided, including attributes and spatial position of a character or object (Stewart et al., 2009). These various results suggest that people integrate information in an ongoing manner, where each incoming word and clause is processed with respect to prior context, and reading time is governed by the computational demands of online evaluative processes.

In humans, the ongoing, word-by-word integration processes might be described as *microinference*, in contrast to *macroinference*, which involves solving complex reasoning and planning problems. In AI, past research on pause tokens has focused on their utility for macroinference (Goyal et al., 2024; Herel and Mikolov, 2023; Pfau et al., 2024; Kim et al., 2025). Our present work is aimed at emphasizing the role of delays for microinference, motivated by the need to integrate and interpret information in an ongoing fashion. One important clue that this focus will be productive comes from a study which found that GPT-2 attention patterns could predict human reading times, in particular measures of the dispersion of attention and the change in attention patterns across time steps (Oh and Schuler, 2022).

## 2 Catch Your Breath

In this section, we describe our approach of dynamically scaling the number of compute steps for individual tokens. We refer reader to Table 1 for notation glossary.

In past research involving pause tokens (see Section 3 for more details), each input token $t$ is granted a certain number of steps, $W_t \in \{1, 2, ...W^{max}\}$, to produce an output, which involves inserting $W_t - 1$ pauses in the input stream. With causal masking, the model can glean no information about the number of upcoming pauses. Consequently, the model must be prepared to respond at each step

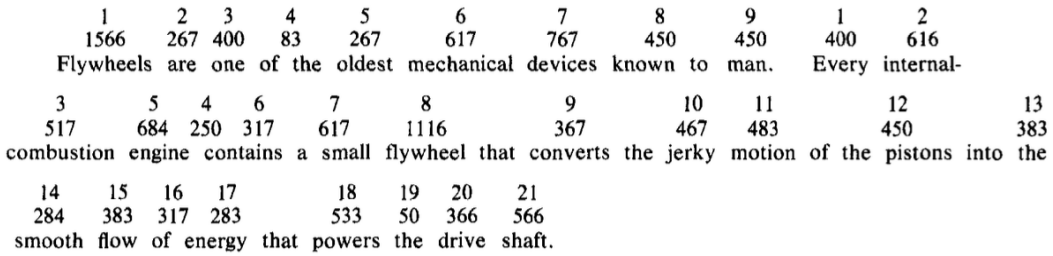


Figure 2: Gaze fixations of a human reader. Indices indicate order of gaze within a sentence along with fixation duration (in msec). Reprinted from Just and Carpenter (1980, Figure 1).

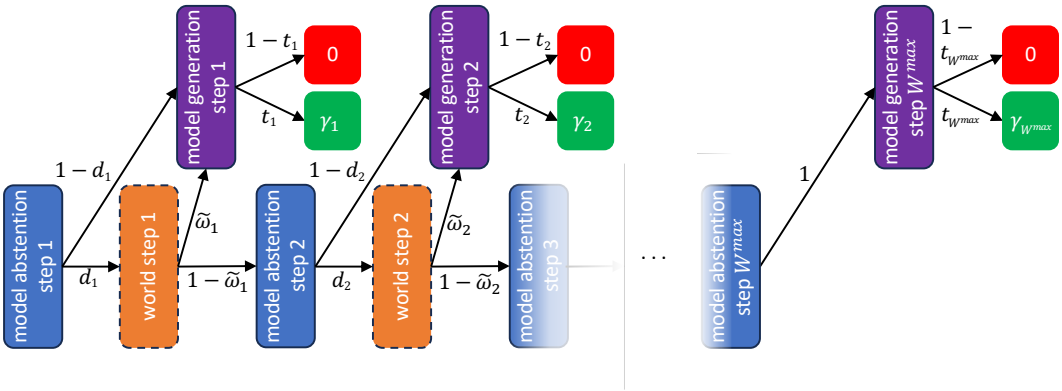


Figure 3: Catch Your Breath: A sequential decision task to obtain a single output token. The blue boxes represent model <DON'T KNOW> decisions, the purple boxes model choices of output tokens. The orange boxes are choices made by the environment about whether the model is granted a delay in responding. The green and red boxes represent end-state rewards.

where these additional pause tokens are inserted, leading to greedy optimization even when additional compute time is allocated.

Because $W_t$ is externally selected, the model cannot anticipate where and how many pauses will occur. From the model's perspective, $W_t$ is thus a random variable. If during training, $W_t$ is drawn consistently in an iid manner from a prior, $W_t \sim \text{Cat}(\boldsymbol{\omega})$, where $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_{W^{max}})$ with $\omega_i = Pr(W_t = i)$, the model's use of pause tokens should be facilitated. However, pause steps might be better utilized if the model itself can control their insertion.

In our approach, we assume that external termination is imposed at some step $W$. (We drop the subscript $t$ since we will always refer to a specific input token.) Additionally, the model can choose to stop sooner: at each step $i < W$, the model can choose either to respond or to request an additional computation step via a <DON'T KNOW> response, chosen with probability $d_i \in [0, 1]$, which reflects the model's uncertainty. If the model chooses to respond or $i = W$, a token is drawn from the usual next-token distribution (excluding <DON'T KNOW>).

Our approach, which we refer to as *Catch Your Breath* or *CYB*, involves framing the selection of each individual output token as a sequential decision problem, depicted in Figure 3. The subscripts in the Figure are a step index relative to the presentation of the real input token. The model has a sequence of decision points, indicated by the blue and purple boxes. At each step $i$, the model selects <DON'T KNOW> with probability $d_i$, a value that is dependent on model parameters and input. When the model is confident of its answer (probability $1 - d_i$), it selects an output token, either the target, with probability $t_i$ or an incorrect response, with probability $1 - t_i$. The green squares indicate the reward (accuracy) when correct, discounted by $\gamma_i \in [0, 1]$ at step $i$, and the red squares the non-reward when incorrect. External termination decision points are depicted by the orange boxes, which represent the action of the world to stop the model at step $i$ given that it hasn't stopped previously, denoted by the